Yield and Preference of Annual Grasses Under Horse Grazing

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Introduction

- In 2013, many legume and grass pastures and hay fields in the upper-Midwest were impacted by winterkill, representing a significant problem for livestock owners.
- Compared to hay, pasture is approximately one third the cost, representing an affordable feed option for many horse owners.
- Winterkill, combined with poor spring growing conditions, left many horse owners with the need for emergency pasture forages.
- Spring planted annual grasses can serve as emergency forage, but have not been evaluated under horse grazing.

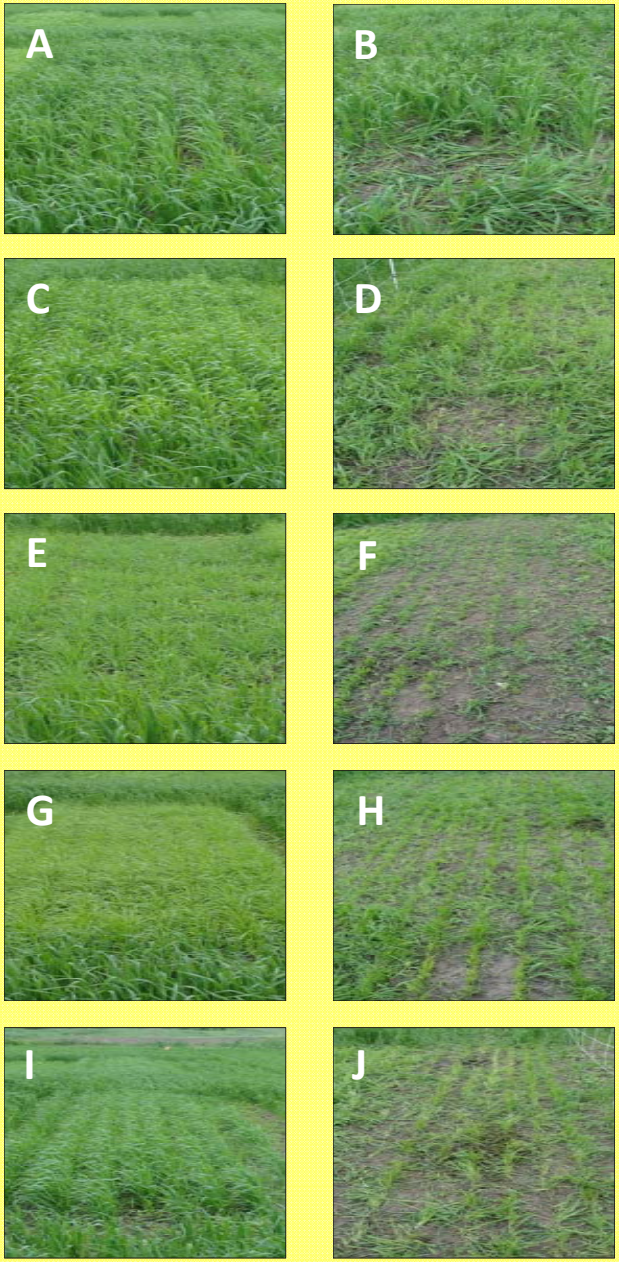
Objective

- The objective of this study was to evaluate yield and preference of spring planted annual grasses under horse grazing.

Materials and Methods

- On May 8, 2013, winter wheat (*Triticum aestivum*), annual ryegrass (*Lolium multiflorum*), barley (*Hordeum vulgare*), spring wheat (*Triticum aestivum*), and oat (*Avena sativa*) were seeded in a randomized complete block design with 8 replicates.
- Each replicate contained the five annual grasses in monocultures.
- Prior to grazing, grass maturity was determined using a scale developed by Moore et al. (1991).
- Yield was determined by hand-harvesting random, duplicate 51 cm^2 areas to 8 cm.
- Samples were weighed and dried for 48 hours in a 60^0C dryer to determine dry matter (DM).
- Three adult horses were acclimated to grazing pastures for 4 hours each day.
- Grazing occurred at two separate maturities; vegetative (June 18, 2013) and reproductive (June 25, 2013).
- At each maturity, horses grazed 4 replicates for 4 hours.
- Immediately after grazing, horse preference was determined by visually assessing percentage of available forage removal on a scale of 0 (no grazing activity) to 100 (100% of the existing vegetation grazed down to 8 cm.).
- Manure was removed and grasses were mowed to 8 cm and allowed to re-grow.

Figure 1. Initial (left) and residual (right) amounts of oat (A,B), barley (C,D), winter wheat (E,F), annual ryegrass (G,H) and spring wheat (I,J) after 4 hours of grazing by 3 adult horses.



Results and Discussion

- There was no difference between the stages of maturities; thus, the two grazing periods were combined ($p \geq 0.0880$).
- Horse preference, determined by percent visual removal, differed among the annual grasses ($P < 0.001$; Table 1).
- Winter wheat and spring wheat were the most preferred grasses with percent visual removal ≥71%.
- Oat was the least preferred with a percent visual removal of 9% ($P < 0.001$).
- Yield also differed among the grasses ($P < 0.001$; Table 1).
- Oat and spring wheat yielded the highest with ≥ 4.8 t ha^{-1} while winter wheat yielded the least at 2.6 t ha^{-1}.

Table 1. Preference and forage yield (on a dry matter basis) of annual grasses under horse grazing.

	Removal ±	Yield
Species	%	t ha^{-1}
Winter Wheat	91^{a*}	2.6^{d}
Spring Wheat	71^{ab}	4.8^{ab}
Barley	56^{b}	4.3^{b}
Annual Ryegrass	55^{b}	3.6^{c}
Oat	9^{c}	5.1^{a}

*Means within a column followed by the same letter are not significant based on Tukey's test ($P < 0.05$).

Conclusions

- Similar to previous research, horses exhibited a strong preference while grazing different species (Allen et al., 2013).
- Spring wheat appears to be both preferred and high yielding and may represent a viable option for horse owners in need of an emergency grazing forage.
- This research is ongoing and will include forage nutritive value of the annual grasses and its impact on horse preference.

References

Allen E., Sheaffer C., Martinson K. 2013. Forage Nutritive Value and Preference of Cool-Season Grasses under Horse Grazing. Agronomy Journal. 105(3). 679-684.

Moore K.J., Moser L.E., Vogel K.P., Waller S.S., Johnson B.E., Pedersen J.F.. 1991. Describing and Quantifying Growth Stages of Perennial Forage Grasses. Agronomy Journal. 83(11). 1073-1077.